CIS Hackathon Documentation:

Title: Patch & Update Risk Management with Risk Analysis

Team Details:

2420080021 – N V Srikar

2420080056 – G Sreeniketh

2420080058 – P Mani Vardhan

2420030305 - P Snehith

Team No: 16

Github URL:

https://github.com/Srikar800/CIS_Hackathon

Details:

This project is a dashboard system that helps organizations:

✅ Track missing software patches
✅ Identify risky systems
✅ Prioritize which systems to update first
✅ Reduce cyberattack risk

In simple words:

👉 It helps companies know which computers/servers are dangerous if not updated.

In companies and cloud systems:

- Many servers need updates regularly
- Some updates are critical security patches
- If updates are missed → hackers can attack

Real example:

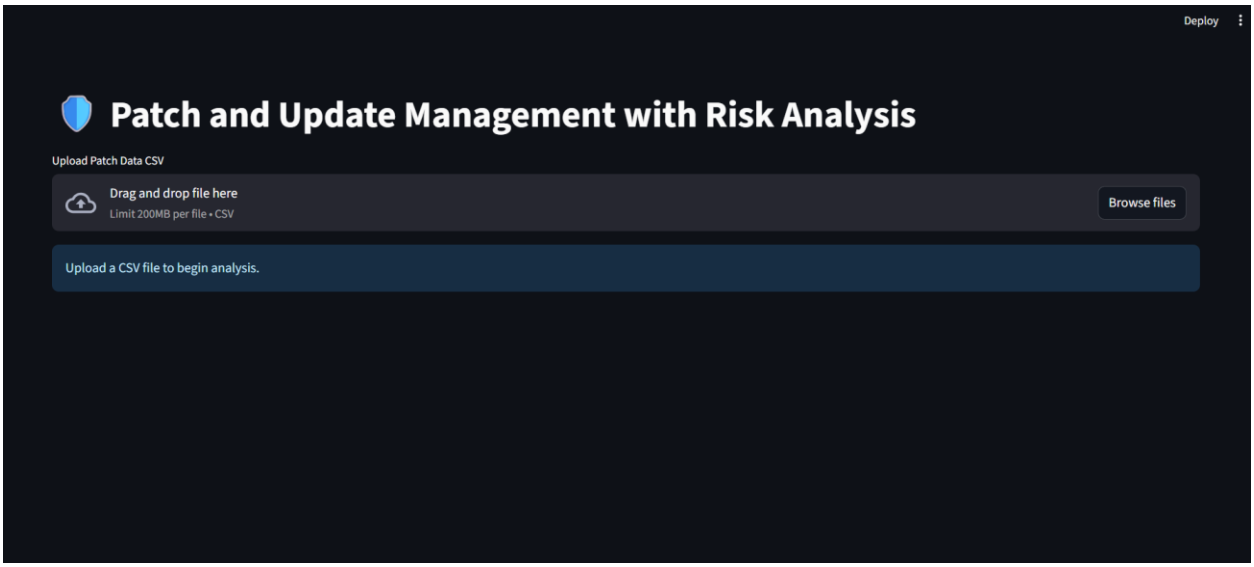
- Missing OpenSSL patch → hackers can steal data
- Unpatched web server → website can be hacked

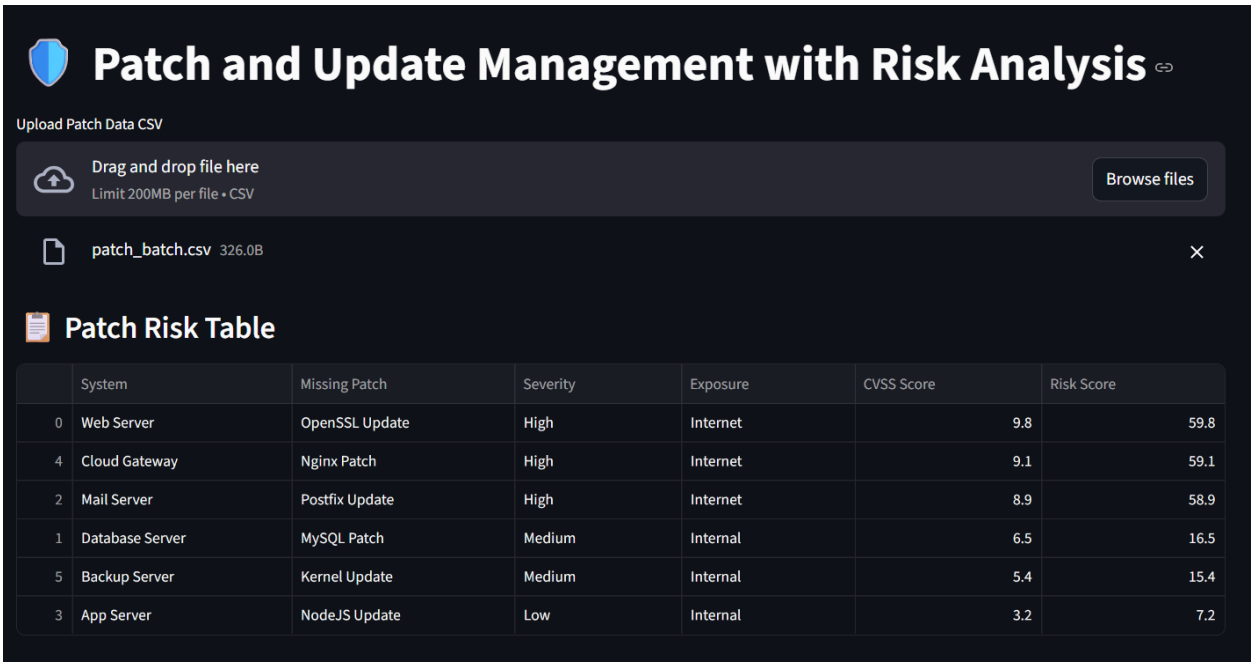
So companies need:

✅ Patch tracking
✅ Risk analysis
✅ Priority management

That’s exactly what your project does.

Proof:





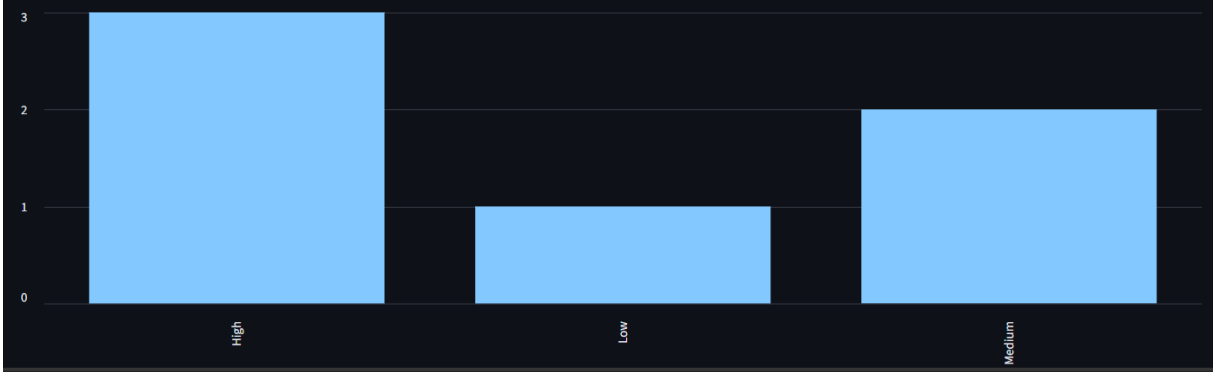
	System	Missing Patch	Severity	Exposure	CVSS Score	Risk Score
0	Web Server	OpenSSL Update	High	Internet	9.8	59.8
4	Cloud Gateway	Nginx Patch	High	Internet	9.1	59.1
2	Mail Server	Postfix Update	High	Internet	8.9	58.9
1	Database Server	MySQL Patch	Medium	Internal	6.5	16.5
5	Backup Server	Kernel Update	Medium	Internal	5.4	15.4
3	App Server	NodeJS Update	Low	Internal	3.2	7.2

High Risk Alerts

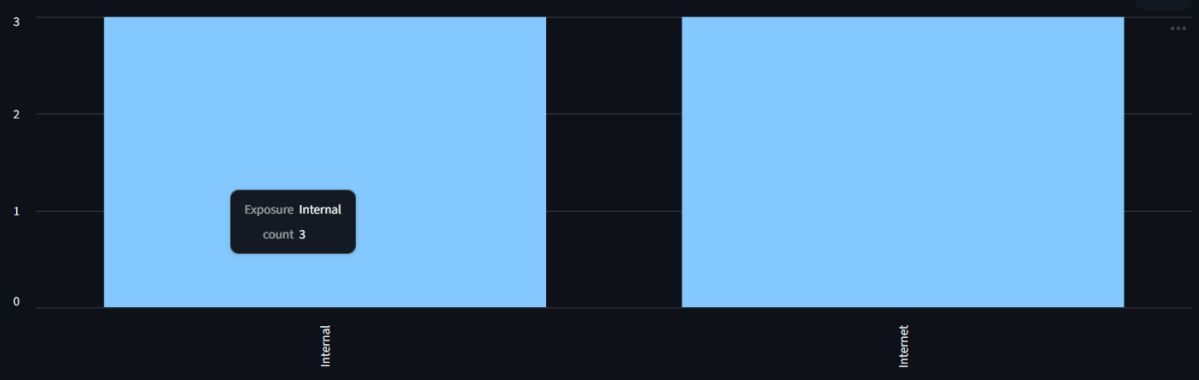
3 High Risk Systems Need Immediate Patching!

	System	Missing Patch	Severity	Exposure	CVSS Score	Risk Score
0	Web Server	OpenSSL Update	High	Internet	9.8	59.8
4	Cloud Gateway	Nginx Patch	High	Internet	9.1	59.1
2	Mail Server	Postfix Update	High	Internet	8.9	58.9

Risk Distribution



Exposure Distribution



Top Priority Patch Systems

	System	Missing Patch	Severity	Exposure	CVSS Score	Risk Score
0	Web Server	OpenSSL Update	High	Internet	9.8000	59.8000
4	Cloud Gateway	Nginx Patch	High	Internet	9.1000	59.1000
2	Mail Server	Postfix Update	High	Internet	8.9000	58.9000
1	Database Server	MySQL Patch	Medium	Internal	6.5000	16.5000
5	Backup Server	Kernel Update	Medium	Internal	5.4000	15.4000